Appendix A -- Sample Apache Configuration httpd.conf

```
############################
### Server configuration ###
############################

###
# Server-wide config
###

# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, in addition to the default. See also the VirtualHost command
Listen localhost:8080

# Load necessary modules which are shared objects.

#
# Dynamic Shared Object (DSO) Support
#
# To be able to use the functionality of a module which was built as a DSO you
# have to place corresponding `LoadModule' lines at this location so the
# directives contained in it are actually available _before_ they are used.
# Statically compiled modules (those listed by `httpd -l') do not need
# to be loaded here.
#
# Example:
# LoadModule foo_module modules/mod_foo.so
#
#LoadModule ssl_module modules/mod_ssl.so
LoadModule rewrite_module modules/mod_rewrite.so
LoadModule jk_module modules/mod_jk_2.0.46.dll
LoadModule access_module modules/mod_access.so
LoadModule actions_module modules/mod_actions.so
LoadModule alias_module modules/mod_alias.so
LoadModule asis_module modules/mod_asis.so
LoadModule auth_module modules/mod_auth.so
LoadModule autoindex_module modules/mod_autoindex.so
LoadModule cgi_module modules/mod_cgi.so
LoadModule dir_module modules/mod_dir.so
LoadModule env_module modules/mod_env.so
LoadModule imap_module modules/mod_imap.so
LoadModule include_module modules/mod_include.so
LoadModule isapi_module modules/mod_isapi.so
LoadModule log_config_module modules/mod_log_config.so
LoadModule mime_module modules/mod_mime.so
LoadModule negotiation_module modules/mod_negotiation.so
LoadModule setenvif_module modules/mod_setenvif.so
LoadModule userdir_module modules/mod_userdir.so

# HostnameLookups: Log the names of clients or just their IP numbers
#   e.g.   www.apache.org (on) or 204.62.129.132 (off)
HostnameLookups off

# ServerAdmin: Your address, where problems with the server should be
# e-mailed.
ServerAdmin garrel@mail.more.net
# ServerName allows you to set a host name which is sent back to clients for
# your server if it's different than the one the program would get (i.e. use
# "www" instead of the host's real name).
#
# Note: You cannot just invent host names and hope they work. The name you
```

```
# define here must be a valid DNS name for your host. If you don't understand
# this, ask your network administrator.
ServerName localhost

# ServerRoot: The directory the server's config, error, and log files
# are kept in

ServerRoot "C:/Program Files/Apache Group/Apache2"
###
# Logging
###
# LogFormat
LogFormat "%h %l %u %t \"%r\" %s %b \"%{Referer}i\" \"%{User-Agent}i\"" combined

# ErrorLog: The location of the error log file. If this does not start
# with /, ServerRoot is prepended to it.
ErrorLog /WebPublish/localhost/logs/error.log

#
# LogLevel: Control the number of messages logged to the error_log.
# Possible values include: debug, info, notice, warn, error, crit,
# alert, emerg.
#
LogLevel warn

#
# The location and format of the access logfile (Common Logfile Format).
# If you do not define any access logfiles within a <VirtualHost>
# container, they will be logged here.  Contrariwise, if you *do*
# define per-<VirtualHost> access logfiles, transactions will be
# logged therein and *not* in this file.
#
CustomLog /WebPublish/localhost/logs/access.log combined


#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (error documents, FTP directory listings,
# mod_status and mod_info output etc., but not CGI generated documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of:  On | Off | EMail
#
ServerSignature Off

# PidFile: The file the server should log its pid to
#PidFile /usr/local/kinetic/apache/logs/httpd.pid
PidFile logs/httpd.pid

# UserDir: The name of the directory which is appended onto a user's home
# directory if a ~user request is recieved.
#UserDir www
#UserDir "My Documents/My Website"

# Config files
TypesConfig conf/mime.types

###
# Server Tuning
###
# CacheNegotiatedDocs: By default, Apache sends Pragma: no-cache with each
# document that was negotiated on the basis of content. This asks proxy
```

```
# servers not to cache the document. Uncommenting the following line disables
# this behavior, and proxies will be allowed to cache the documents.

#CacheNegotiatedDocs

#
# UseCanonicalName:  (new for 1.3)  With this setting turned on, whenever
# Apache needs to construct a self-referencing URL (a URL that refers back
# to the server the response is coming from) it will use ServerName and
# Port to form a "canonical" name.  With this setting off, Apache will
# use the hostname:port that the client supplied, when possible.  This
# also affects SERVER_NAME and SERVER_PORT in CGI scripts.
#
UseCanonicalName On

# Timeout: The number of seconds before receives and sends time out
#  n.b. the compiled default is 1200 (20 minutes !)
Timeout 180

# KeepAlive: The number of Keep-Alive persistent requests to accept
# per connection. Set to 0 to deactivate Keep-Alive support
KeepAlive On

#
# MaxKeepAliveRequests: The maximum number of requests to allow
# during a persistent connection. Set to 0 to allow an unlimited amount.
# We recommend you leave this number high, for maximum performance.
#
MaxKeepAliveRequests 100

# KeepAliveTimeout: Number of seconds to wait for the next request
KeepAliveTimeout 2

###
# Server Resources
###

# DocumentRoot: The directory out of which you will serve your
# documents. By default, all requests are taken from this directory, but
# symbolic links and aliases may be used to point to other locations.
DocumentRoot "/WebPublish/localhost/www"

# DirectoryIndex: Name of the file or files to use as a pre-written HTML
# directory index.  Separate multiple entries with spaces.
DirectoryIndex index.html index.htm index.shtml index.shtm

# FancyIndexing is whether you want fancy directory indexing or standard
# IndexOptions FancyIndexing

###
# Aliases & Icons
###
# Alias fakename realname
Alias /icons/ "C:/Program Files/Apache Group/Apache2/icons/"

<Directory "C:/Program Files/Apache Group/Apache2/icons">
    Options Indexes MultiViews
    AllowOverride None
    Order allow,deny
    Allow from all
</Directory>
```

```
# AddIcon tells the server which icon to show for different files or filename
AddIconByEncoding (CMP,/icons/compressed.gif) x-compress x-gzip
AddIconByType (TXT,/icons/text.gif) text/*
AddIconByType (IMG,/icons/image2.gif) image/*
AddIconByType (SND,/icons/sound2.gif) audio/*
AddIconByType (VID,/icons/movie.gif) video/*
AddIcon /icons/binary.gif .bin .exe
AddIcon /icons/binhex.gif .hqx
AddIcon /icons/tar.gif .tar
AddIcon /icons/world2.gif .wrl .wrl.gz .vrml .vrm .iv
AddIcon /icons/compressed.gif .Z .z .tgz .gz .zip
AddIcon /icons/a.gif .ps .ai .eps
AddIcon /icons/layout.gif .html .shtml .htm .pdf
AddIcon /icons/text.gif .txt
AddIcon /icons/c.gif .c
AddIcon /icons/p.gif .pl .py
AddIcon /icons/f.gif .for
AddIcon /icons/dvi.gif .dvi
AddIcon /icons/uuencoded.gif .uu
AddIcon /icons/script.gif .conf .sh .shar .csh .ksh .tcl
AddIcon /icons/tex.gif .tex
AddIcon /icons/bomb.gif core
AddIcon /icons/back.gif ..
AddIcon /icons/hand.right.gif README
AddIcon /icons/folder.gif ^^DIRECTORY^^
AddIcon /icons/blank.gif ^^BLANKICON^^

# DefaultIcon is which icon to show for files which do not have an icon
# explicitly set.
DefaultIcon /icons/unknown.gif

# AddDescription allows you to place a short description after a file in
# server-generated indexes.
# Format: AddDescription "description" filename

# ReadmeName is the name of the README file the server will look for by
# default. Format: ReadmeName name
#
# The server will first look for name.html, include it if found, and it will
# then look for name and include it as plaintext if found.
#
# HeaderName is the name of a file which should be prepended to
# directory indexes.
ReadmeName README
HeaderName HEADER

IndexIgnore .??* *~ *# HEADER* README* RCS CVS *,v *,t WEB-INF META-INF

AccessFileName .htaccess

#
# The following lines prevent .htaccess files from being viewed by
# Web clients.  Since .htaccess files often contain authorization
# information, access is disallowed for security reasons.  Comment
# these lines out if you want Web visitors to see the contents of
# .htaccess files.  If you change the AccessFileName directive above,
# be sure to make the corresponding changes here.
#
<Files .htaccess>
    Order allow,deny
```

```
    Deny from all
</Files>

# DefaultType is the default MIME type for documents which the server
# cannot find the type of from filename extensions.
DefaultType text/plain

# AddEncoding allows you to have certain browsers (Mosaic/X 2.1+) uncompress
# information on the fly. Note: Not all browsers support this.
AddEncoding x-compress Z
AddEncoding x-gzip gz

# AddLanguage allows you to specify the language of a document. You can
# then use content negotiation to give a browser a file in a language
# it can understand.  Note that the suffix does not have to be the same
# as the language keyword --- those with documents in Polish (whose
# net-standard language code is pl) may wish to use "AddLanguage pl .po"
# to avoid the ambiguity with the common suffix for perl scripts.
AddLanguage en .en
AddLanguage fr .fr
AddLanguage de .de
AddLanguage da .da
AddLanguage el .el
AddLanguage it .it

# LanguagePriority allows you to give precedence to some languages
# in case of a tie during content negotiation.
# Just list the languages in decreasing order of preference.
LanguagePriority en fr de

# AddType allows you to tweak mime.types without actually editing it, or to
# make certain files to be certain types.
# Format: AddType type/subtype ext1

# AddHandler allows you to map certain file extensions to "handlers",
# actions unrelated to filetype. These can be either built into the server
# or added with the Action command (see below)
# Format: AddHandler action-name ext1

# To use server-parsed HTML files
AddType text/html .shtml .shtm .html .htm
AddType application/octet-stream doc
AddType application/octet-stream dot
AddType application/octet-stream exe
AddType application/octet-stream ppt
AddType application/octet-stream sea
AddType application/octet-stream srm
AddType application/octet-stream xls
AddType application/octet-stream mdb
AddType application/octet-stream wpd wp5 wp
AddType application/x-director dcr dir dxr
AddType image/x-Freehand fhc fh4 fh5
AddType audio/x-pn-realaudio raplayer  rm ra ram
AddType audio/x-pn-real audio-plugin    rpm

<Location />
AddOutputFilter INCLUDES .shtml .shtm
</Location>

# If you wish to use server-parsed imagemap files, use
AddHandler imap-file map
```

```
# To enable type maps, you might want to use
AddHandler type-map var

# Tomcat mod_jk configuration

JkWorkersFile "C:/Program Files/Apache Group/Apache2/conf/workers.properties"
JkLogFile "/WebPublish/localhost/logs/mod_jk.log"
JkLogLevel error
JkLogStampFormat "[%a %b %d %H:%M:%S %Y] "

#
# The following directives modify normal HTTP response behavior.
# The first directive disables keepalive for Netscape 2.x and browsers that
# spoof it. There are known problems with these browser implementations.
# The second directive is for Microsoft Internet Explorer 4.0b2
# which has a broken HTTP/1.1 implementation and does not properly
# support keepalive when it is used on 301 or 302 (redirect) responses.
#
BrowserMatch "Mozilla/2" nokeepalive
BrowserMatch "MSIE 4\.0b2;" nokeepalive downgrade-1.0 force-response-1.0

#
# The following directive disables HTTP/1.1 responses to browsers which
# are in violation of the HTTP/1.0 spec by not being able to grok a
# basic 1.1 response.
#
BrowserMatch "RealPlayer 4\.0" force-response-1.0
BrowserMatch "Java/1\.0" force-response-1.0
BrowserMatch "JDK/1\.0" force-response-1.0

#
# Each directory to which Apache has access, can be configured with respect
# to which services and features are allowed and/or disabled in that
# directory (and its subdirectories).
#
# First, we configure the "default" to be a very restrictive set of
# permissions.
#
<Directory />
    Options FollowSymLinks
    AllowOverride None
</Directory>

#
# Global settings for documents in web space
#
<Directory /WebPublish/localhost/www>
Options Indexes Includes FollowSymLinks
AllowOverride AuthConfig FileInfo Indexes Limit
</Directory>


<VirtualHost localhost:8080>
ServerAdmin garrel@mail.more.net
DocumentRoot "/WebPublish/localhost/www"
ServerName localhost

DirectoryIndex index.html index.htm index.shtml index.shtm index.jsp

JkRequestLogFormat "%w %v \"%r\" \"%U%q\" %s %T"
```

```
JkAutoAlias "/WebPublish/localhost/www"
JkMount /*.jsp tomcat
JkMount /*/servlet/ tomcat
JkMount /*.do tomcat
JkMount /manager/* tomcat


<LocationMatch ".*\.(do|jsp)$">
Options Includes
AddOutputFilterByType INCLUDES text/html
</LocationMatch>

<LocationMatch "^/[^\/]+/servlet/">
Options Includes
AddOutputFilterByType INCLUDES text/html
</LocationMatch>

<Location /ethemesrequest/ >
Options Includes
AddOutputFilterByType INCLUDES text/html
</Location>

</VirtualHost>

<LocationMatch "/WEB-INF/">
    AllowOverride None
    deny from all
</LocationMatch>
```

Appendix B -- Sample Tomcat Configuration server.xml

```
<!-- Example Server Configuration File -->
<!-- Note that component elements are nested corresponding to their
     parent-child relationships with each other -->

<!-- A "Server" is a singleton element that represents the entire JVM,
     which may contain one or more "Service" instances.  The Server
     listens for a shutdown command on the indicated port.

     Note:  A "Server" is not itself a "Container", so you may not
     define subcomponents such as "Valves" or "Loggers" at this level.
 -->

<Server port="8005" shutdown="SHUTDOWN" debug="0">

  <!-- Uncomment these entries to enable JMX MBeans support
  <Listener className="org.apache.catalina.mbeans.ServerLifecycleListener"
            debug="0"/>
  <Listener
    className="org.apache.catalina.mbeans.GlobalResourcesLifecycleListener"
    debug="0"/>
  -->


  <!-- A "Service" is a collection of one or more "Connectors" that share
       a single "Container" (and therefore the web applications visible
       within that Container).  Normally, that Container is an "Engine",
       but this is not required.

       Note:  A "Service" is not itself a "Container", so you may not
       define subcomponents such as "Valves" or "Loggers" at this level.
   -->

  <!-- Define the Service for Tomcat and Apache -->
  <Service name="Tomcat-Apache">

    <!-- A "Connector" represents an endpoint by which requests are received
         and responses are returned.  Each Connector passes requests on to the
         associated "Container" (normally an Engine) for processing.

         By default, a non-SSL HTTP/1.1 Connector is established on port 8080.
         You can also enable an SSL HTTP/1.1 Connector on port 8443 by
         following the instructions below and uncommenting the second Connector
         entry.  SSL support requires the following steps:
         * Download and install JSSE 1.0.2 or later, and put the JAR files
           into "$JAVA_HOME/jre/lib/ext".
         * Edit "$JAVA_HOME/jre/lib/security/java.security" and add
             security.provider.2=com.sun.net.ssl.internal.ssl.Provider
         * Execute: keytool -genkey -alias tomcat -keyalg RSA
           with a password value of "changeit".

         By default, DNS lookups are enabled when a web application calls
         request.getRemoteHost().  This can have an adverse impact on
         performance, so you can disable it by setting the
         "enableLookups" attribute to "false".  When DNS lookups are disabled,
         request.getRemoteHost() will return the String version of the
         IP address of the remote client.
    -->

    <!-- Define an AJP13 Connector to Apache on port 8009 -->
```

```
<Connector className="org.apache.ajp.tomcat4.Ajp13Connector"
           port="8009" minProcessors="5" maxProcessors="75"
           acceptCount="10" connectionTimeout="0" debug="0"/>

<!-- An Engine represents the entry point (within Catalina) that processes
     every request.  The Engine implementation for Tomcat stand alone
     analyzes the HTTP headers included with the request, and passes them
     on to the appropriate Host (virtual host). -->

<!-- Define the top level container in our container hierarchy -->
<Engine name="Tomcat-Apache" defaultHost="localhost" debug="0">

  <!-- The request dumper valve dumps useful debugging information about
       the request headers and cookies that were received, and the response
       headers and cookies that were sent, for all requests received by
       this instance of Tomcat.  If you care only about requests to a
       particular virtual host, or a particular application, nest this
       element inside the corresponding <Host> or <Context> entry instead.

       For a similar mechanism that is portable to all Servlet 2.3
       containers, check out the "RequestDumperFilter" Filter in the
       example application (the source for this filter may be found in
       "$CATALINA_HOME/webapps/examples/WEB-INF/classes/filters").

       Request dumping is disabled by default.  Uncomment the following
       element to enable it. -->
  <!--
  <Valve className="org.apache.catalina.valves.RequestDumperValve"/>
  -->

  <!-- Global logger unless overridden at lower levels -->
  <Logger className="org.apache.catalina.logger.FileLogger"
          prefix="catalina_log." suffix=".txt"
          timestamp="true"
          directory="C:\WebPublish\localhost\tomcat\logs" />


  <!-- Define the default virtual host -->
  <Host name="localhost" debug="0"
        appBase="C:\WebPublish\localhost\www"
        workDir="C:\WebPublish\localhost\tomcat\work"
        unpackWARs="true" swallowOutput="true"
        autoDeploy="true" liveDeploy="false" deployXML="false">

    <!-- realm for Tomcat Manager authentication using MySQL database -->
    <Realm className="org.apache.catalina.realm.JDBCRealm" debug="0"
           driverName="com.mysql.jdbc.Driver"
           connectionURL="jdbc:mysql://localhost/local_realm?autoReconnect=true"
           connectionName="localrealm"
           connectionPassword="xxxxxxxx"
           userTable="users" userNameCol="user_name" userCredCol="user_pass"
           userRoleTable="user_roles" roleNameCol="role_name" digest="MD5"/>

    <!-- Logger shared by all Contexts related to this virtual host.  By
         default (when using FileLogger), log files are created in the "logs"
         directory relative to $CATALINA_HOME.  If you wish, you can specify
         a different directory with the "directory" attribute.  Specify either a
         relative (to $CATALINA_HOME) or absolute path to the desired
         directory.-->
    <Logger className="org.apache.catalina.logger.FileLogger"
            directory="C:\WebPublish\localhost\tomcat\logs"
```

```
        prefix="localhost_log." suffix=".txt"
       timestamp="true"/>

<!-- Define properties for each web application.  This is only needed
     if you want to set non-default properties, or have web application
     document roots in places other than the virtual host's appBase
     directory.  -->

<DefaultContext debug="0" reloadable="false" swallowOutput="true">
  <!-- JNDI Resource for sending email using SMTP -->
  <Resource name="mail/send" auth="CONTAINER"
            type="javax.mail.internet.MimePartDataSource"/>
  <ResourceParams name="mail/send">
    <parameter><name>factory</name>
      <value>org.apache.naming.factory.SendMailFactory</value>
    </parameter>
    <parameter>
      <name>mail.smtp.host</name>

      <value>yourhost.yourdomain</value>
    </parameter>
    <parameter>
      <name>mail.smtp.user</name>
      <value>your-userid</value>
    </parameter>
  <parameter>
      <name>mail.from</name>
      <value>your-email-address</value>
    </parameter>
  </ResourceParams>
  <!-- JNDI JDBC DataSource Resource for using MySQL dB -->
  <Resource name="jdbc/data" auth="CONTAINER"
            type="javax.sql.DataSource"/>
  <ResourceParams name="jdbc/data">
    <parameter>
      <name>factory</name>
      <value>org.apache.commons.dbcp.BasicDataSourceFactory</value>
    </parameter>
    <parameter>
      <name>username</name>
      <value>localweb</value>
    </parameter>
    <parameter>
      <name>password</name>
      <value>your-password-for-db-access-in-MySQL</value>
    </parameter>
    <parameter>
      <name>driverClassName</name>
      <value>com.mysql.jdbc.Driver</value></parameter>
    <parameter>
      <name>url</name>
      <value>jdbc:mysql://localhost/local?autoReconnect=true</value>
    </parameter>
    <parameter>
      <name>maxIdle</name>
      <value>10</value>
    </parameter>
    <parameter>
      <name>maxActive</name>
      <value>50</value>
    </parameter>
```

```
            <parameter>
              <name>maxWait</name>
              <value>10000</value>
            </parameter>
            <parameter>
              <name>logAbandoned</name>
              <value>true</value>
            </parameter>
            <parameter>
              <name>removeAbandoned</name>
              <value>true</value>
            </parameter>
            <parameter>
              <name>removeAbandonedTimeout</name>
              <value>300</value>
            </parameter>
          </ResourceParams>
        </DefaultContext>

        <!-- Tomcat Manager Context -->
        <Context path="/manager"
                 docBase="C:\Program Files\Apache Group\Tomcat 4.1\server\webapps\manager"
                 debug="0" privileged="true">
          <Valve className="org.apache.catalina.valves.RemoteAddrValve"
                 allow="127.0.0.1" />
        </Context>



      </Host>

    </Engine>

  </Service>

</Server>
```

Appendix C -- Sample Tomcat Security Policy catalina.policy

```
// ============================================================================
// catalina.corepolicy - Security Policy Permissions for Tomcat 4.1
//
// This file contains a default set of security policies to be enforced (by the
// JVM) when Catalina is executed with the "-security" option.  In addition
// to the permissions granted here, the following additional permissions are
// granted to the codebase specific to each web application:
//
// * Read access to the document root directory
//
// ============================================================================


// ========== SYSTEM CODE PERMISSIONS =========================================


// These permissions apply to javac
grant codeBase "file:${java.home}/../lib/tools.jar" {
  permission java.security.AllPermission;
};

// These permissions apply to all shared system extensions
grant codeBase "file:${java.home}/lib/-" {
  permission java.security.AllPermission;
};


// ========== CATALINA CODE PERMISSIONS =======================================

// These permissions apply to the server startup code, and the servlet API
// classes that are shared across all class loaders
grant codeBase "file:${catalina.home}/bin/bootstrap.jar" {
  // Tomcat should be able to read/write all properties
  permission java.util.PropertyPermission "*", "read,write";
  // Tomcat needs to be able to read java resources
  permission java.io.FilePermission "${java.home}${/}..${/}-", "read";
  // Jasper needs permission for ant to execute javac
  permission java.io.FilePermission "${java.home}${/}..${/}bin${/}javac.exe", "execute";
  // Tomcat needs to be able to read files in its own directory
  permission java.io.FilePermission "${catalina.home}", "read";
  permission java.io.FilePermission "${catalina.home}${/}-", "read";
  // Tomcat needs to be able to unarchive war files and write the files
  permission java.io.FilePermission
    "C:${/}WebPublish${/}localhost${/}-", "read,write,delete";
  permission java.io.FilePermission "${catalina.home}${/}webapps${/}-", "write,delete";
  // Tomcat has to be able to write its logs
  permission java.io.FilePermission "${catalina.home}${/}logs${/}-", "write";
  // Tomcat has to be able to compile JSP's
  permission java.io.FilePermission "${catalina.home}${/}work${/}-", "write,delete";
  // Tomcat needs permission to read/write Windows temp directory
  permission java.io.FilePermission "${catalina.home}${/}temp${/}-", "write,delete";
  // Tomcat admin needs permission to update config files
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.new", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.old", "write,delete";
  // Required for reading resources using JNDI lookup
```

```
  permission org.apache.naming.JndiPermission "*";
  // Tomcat needs permission to do RuntimePermissions like create
  // a ClassLoader and exit the VM.
  permission java.lang.RuntimePermission "*";
  // Tomcat needs permission to do reflection
  permission java.lang.reflect.ReflectPermission "*";
  // Needed so Tomcat can set security policy for a Context
  permission java.security.SecurityPermission "*";
  // Tomcat needs these permissions for JNDI resources
  permission java.net.NetPermission "*";
  // Permission to connect to the MySQL database server
  permission java.net.SocketPermission "localhost:3306", "connect";
  // Tomcat will allow outgoing HTTP requests
  permission java.net.SocketPermission "*:80", "connect";
  // Tomcat has to be able to use its port on the localhost
  permission java.net.SocketPermission "localhost", "connect,accept";
  permission java.net.SocketPermission "localhost:8005", "listen";
  permission java.net.SocketPermission "localhost:8080", "listen";
  permission java.net.SocketPermission "localhost:8009", "listen";
};

// These permissions apply to the container's core code, plus any additional
// libraries installed in the "server" directory
grant codeBase "file:${catalina.home}/server/-" {
  // Tomcat should be able to read/write all properties
  permission java.util.PropertyPermission "*", "read,write";
  // Tomcat needs to be able to read java resources
  permission java.io.FilePermission "${java.home}${/}..${/}-", "read";
  // Tomcat needs to be able to read files in its own directory
  permission java.io.FilePermission "${catalina.home}", "read";
  permission java.io.FilePermission "${catalina.home}${/}-", "read";
  // Tomcat needs to be able to unarchive war files and write the files
  permission java.io.FilePermission
    "C:${/}WebPublish${/}localhost${/}-", "read,write,delete";
  permission java.io.FilePermission "${catalina.home}${/}webapps${/}-", "write,delete";
  // Tomcat has to be able to write its logs
  permission java.io.FilePermission "${catalina.home}${/}logs${/}-", "write";
  // Tomcat has to be able to compile JSP's
  permission java.io.FilePermission "${catalina.home}${/}work${/}-", "write,delete";
  // Tomcat needs permission to read/write Windows temp directory
  permission java.io.FilePermission "${catalina.home}${/}temp${/}-", "write,delete";
  // Tomcat admin needs permission to update config files
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.new", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.old", "write,delete";
  // Jasper needs permission for ant to execute javac
  permission java.io.FilePermission "${java.home}${/}..${/}bin${/}javac.exe", "execute";
  // Required for reading resources using JNDI lookup
  permission org.apache.naming.JndiPermission "*";
  // Tomcat needs permission to do RuntimePermissions like create
  // a ClassLoader and exit the VM.
  permission java.lang.RuntimePermission "*";
  // Tomcat needs permission to do reflection
  permission java.lang.reflect.ReflectPermission "*";
  // Needed so Tomcat can set security policy for a Context
  permission java.security.SecurityPermission "*";
  // Tomcat needs these permissions for JNDI resources
  permission java.net.NetPermission "*";
```

```
  // Permission to connect to the MySQL database server
  permission java.net.SocketPermission "localhost:3306", "connect";
  // Tomcat will allow outgoing HTTP requests
  permission java.net.SocketPermission "*:80", "connect";
  // Tomcat has to be able to use its port on the localhost
  permission java.net.SocketPermission "localhost", "connect,accept";
  permission java.net.SocketPermission "localhost:8005", "listen";
  permission java.net.SocketPermission "localhost:8080", "listen";
  permission java.net.SocketPermission "localhost:8009", "listen";
};

// These permissions are for the classes used for JNDI
grant codeBase "file:${catalina.home}/common/lib/naming-factory.jar" {
  // For the UserDatabase Realm
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.catalina.*";
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.catalina.*";
  // Tomcat admin needs permission to update config files
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.new", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.old", "write,delete";
};

// These permissions are for the classes used for JNDI
grant codeBase "file:${catalina.home}/common/lib/naming-common.jar" {
  // For the UserDatabase Realm
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.catalina.*";
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.catalina.*";
  // Tomcat admin needs permission to update config files
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.new", "write";
  permission java.io.FilePermission
    "${catalina.home}${/}conf${/}tomcat-users.xml.old", "write,delete";
};

// These permissions are for the classes used for compiling JSP pages
grant codeBase "file:${catalina.home}/common/lib/jasper-compiler.jar" {
  // Jasper should be able to read/write all properties
  permission java.util.PropertyPermission "*", "read,write";
  // Jasper needs to be able to read java resources
  permission java.io.FilePermission "${java.home}${/}..${/}-", "read";
  // Jasper needs to be able to read files in its own directory
  permission java.io.FilePermission "${catalina.home}", "read";
  permission java.io.FilePermission "${catalina.home}${/}-", "read";
  // Jasper needs to be able to unarchive war files and write the files
  permission java.io.FilePermission
     "C:${/}WebPublish${/}localhost${/}-", "read,write,delete";
  permission java.io.FilePermission
     "${catalina.home}${/}webapps${/}-", "write,delete";
  // Jasper has to be able to write its logs
  permission java.io.FilePermission "${catalina.home}${/}logs${/}-", "write";
  // Jasper has to be able to compile JSP's
  permission java.io.FilePermission "${catalina.home}${/}work${/}-", "write,delete";
  // Jasper needs permission to read/write Windows temp directory
  permission java.io.FilePermission "${catalina.home}${/}temp${/}-", "write,delete";
  // Required for reading resources using JNDI lookup
  permission org.apache.naming.JndiPermission "*";
```

```
  // Jasper needs permision to access its own package
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.jasper.*";
  // Jasper needs permision to define sun.tools for compiling JSP pages using javac
  permission java.lang.RuntimePermission "defineClassInPackage.sun.*";
  // Jasper needs permision to define its own package
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.jasper.*";
  // Jasper needs permision to access Tomcat utility resources
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.catalina.*";
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.coyote.*";
  // Jasper needs permission to get a ClassLoader
  permission java.lang.RuntimePermission "getClassLoader";
  // Jasper needs permission to create a ClassLoader for a JSP page
  permission java.lang.RuntimePermission "createClassLoader";
  permission java.lang.RuntimePermission "setContextClassLoader";
  permission java.lang.RuntimePermission "setIO";
  // Needed so Jasper can get the security policy for a Context
  permission java.security.SecurityPermission "*";
  // Required by the admin application
  permission java.lang.RuntimePermission "getAttribute";
  permission java.lang.RuntimePermission "setAttribute";
  // Jasper needs permission for ant to execute javac
  permission java.io.FilePermission "${java.home}${/}..${/}bin${/}javac.exe", "execute";
  permission java.lang.RuntimePermission "shutdownHooks";
  // Permission to connect to the MySQL database server
  permission java.net.SocketPermission "localhost:3306", "connect";
};

// These permissions are for the classes used for compiling JSP pages
grant codeBase "file:${catalina.home}/common/lib/ant.jar" {
  // Jasper should be able to read/write all properties
  permission java.util.PropertyPermission "*", "read,write";
  // Jasper needs to be able to read java resources
  permission java.io.FilePermission "${java.home}${/}..${/}-", "read";
  // Jasper needs to be able to read files in its own directory
  permission java.io.FilePermission "${catalina.home}", "read";
  permission java.io.FilePermission "${catalina.home}${/}-", "read";
  // Jasper needs to be able to unarchive war files and write the files
  permission java.io.FilePermission
    "C:${/}WebPublish${/}localhost${/}-", "read,write,delete";
  permission java.io.FilePermission "${catalina.home}${/}webapps${/}-", "write,delete";
  // Jasper has to be able to write its logs
  permission java.io.FilePermission "${catalina.home}${/}logs${/}-", "write";
  // Jasper has to be able to compile JSP's
  permission java.io.FilePermission "${catalina.home}${/}work${/}-", "write,delete";
  // Jasper needs permission to read/write Windows temp directory
  permission java.io.FilePermission "${catalina.home}${/}temp${/}-", "write,delete";
  // Required for reading resources using JNDI lookup
  permission org.apache.naming.JndiPermission "*";
  // Jasper needs permision to access its own package
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.jasper.*";
  // Jasper needs permision to define sun.tools for compiling JSP pages using javac
  permission java.lang.RuntimePermission "defineClassInPackage.sun.*";
  // Jasper needs permision to define its own package
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.jasper.*";
  // Jasper needs permision to access Tomcat utility resources
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.catalina.*";
  // Jasper needs permission to get a ClassLoader
  permission java.lang.RuntimePermission "getClassLoader";
  // Jasper needs permission to create a ClassLoader for a JSP page
  permission java.lang.RuntimePermission "createClassLoader";
  permission java.lang.RuntimePermission "setContextClassLoader";
```

```
  permission java.lang.RuntimePermission "setIO";
  // Needed so Jasper can get the security policy for a Context
  permission java.security.SecurityPermission "*";
  // Jasper needs permission for ant to execute javac
  permission java.io.FilePermission "${java.home}${/}..${/}bin${/}javac.exe", "execute";
  permission java.lang.RuntimePermission "shutdownHooks";
};

// These permissions are for all classes shared by both Tomcat and web applications
grant codeBase "file:${catalina.home}/common/-" {
  // Tomcat should be able to read/write all properties
  permission java.util.PropertyPermission "*", "read,write";
  // Tomcat needs to be able to read java resources
  permission java.io.FilePermission "${java.home}${/}..${/}-", "read";
  // Tomcat needs to be able to read files in its own directory
  permission java.io.FilePermission "${catalina.home}", "read";
  permission java.io.FilePermission "${catalina.home}${/}-", "read";
  // Tomcat needs to be able to unarchive war files and write the files
  permission java.io.FilePermission
    "C:${/}WebPublish${/}localhost${/}-", "read,write,delete";
  permission java.io.FilePermission "${catalina.home}${/}webapps${/}-", "write,delete";
  // Tomcat has to be able to write its logs
  permission java.io.FilePermission "${catalina.home}${/}logs${/}-", "write";
  // Tomcat has to be able to compile JSP's
  permission java.io.FilePermission "${catalina.home}${/}work${/}-", "write,delete";
  // Tomcat needs permission to read/write Windows temp directory
  permission java.io.FilePermission "${catalina.home}${/}temp${/}-", "write,delete";
  // Required for reading resources using JNDI lookup
  permission java.lang.RuntimePermission "getClassLoader";
  // Permission to connect to the MySQL database server
  permission java.net.SocketPermission "localhost:3306", "connect";
};

// These permissions are for JSP pages and servlets
// This sets the permissions that a web application could be granted.
grant codeBase "file:${catalina.home}/common/lib/servlet.jar" {
  // Servlets and JSP pages need permision to access Tomcat utility resources
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.catalina.*";
  // Servlets and JSP pages need permision to define Tomcat utility resources
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.catalina.*";
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.coyote.*";
  // Servlets and JSP pages need permision to access its own package
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.jasper.*";
  // Servlets and JSP pages need permision to define its own package
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.jasper.*";
  // Servlets and JSP pages need permision to define classes in package
  // sun. for compiling JSP pages
  permission java.lang.RuntimePermission "defineClassInPackage.sun.*";
  // Servlets and JSP pages need permission to create a ClassLoader for a JSP page
  permission java.lang.RuntimePermission "createClassLoader";
  permission java.lang.RuntimePermission "setContextClassLoader";
  permission java.lang.RuntimePermission "setIO";
  // Needed so the manager servlet can get the security policy for a
  // Context when installing it.
  permission java.security.SecurityPermission "*";
  // Web applications need this permissions for JNDI resources
  permission org.apache.naming.JndiPermission "*";
  // Required by the admin application
  permission java.lang.RuntimePermission "getAttribute";
  permission java.lang.RuntimePermission "setAttribute";
  // Tomcat needs to be able to read files in its own directory
```

```
  permission java.io.FilePermission "${catalina.home}", "read";
  // Jasper needs permission for ant to execute javac
  permission java.io.FilePermission "${java.home}${/}..${/}bin${/}javac.exe", "execute";
  permission java.lang.RuntimePermission "shutdownHooks";
  // Servlets needs to be able to read, write, and delete web application files
  permission java.io.FilePermission
    "C:${/}WebPublish${/}localhost${/}-", "read,write,delete";
  // Permission to connect to the MySQL database server
  permission java.net.SocketPermission "localhost:3306", "connect";
};

// These permissions apply to the Jasper runtime if present in the common lib directory
grant codeBase "file:${catalina.home}/common/lib/jasper-runtime.jar" {
  // Jasper needs permision to access its own package
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.jasper.*";
  // Jasper needs permision to access Tomcat utility resources
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.catalina.*";
  // Jasper needs permision to define Tomcat utility resources
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.catalina.*";
  // Jasper needs permision to define its own package
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.jasper.*";
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.coyote.*";
  permission java.util.PropertyPermission "*", "read";
  // Jasper needs to be able to read files in its Tomcat's home directory
  permission java.io.FilePermission "${catalina.home}", "read";
  permission java.io.FilePermission "${catalina.home}${/}-", "read";
  // Required by the admin application
  permission java.lang.RuntimePermission "getAttribute";
  permission java.lang.RuntimePermission "setAttribute";
  // Jasper needs permission for ant to execute javac
  permission java.io.FilePermission "${java.home}${/}..${/}bin${/}javac.exe", "execute";
  permission java.lang.RuntimePermission "shutdownHooks";
  // Servlets needs to be able to read, write, and delete web application files
  permission java.io.FilePermission
    "C:${/}WebPublish${/}localhost${/}-", "read,write,delete";
  // Permission to connect to the MySQL database server
  permission java.net.SocketPermission "localhost:3306", "connect";
};

// These permissions apply to the Xerces runtime if present in the common
// endorsed directory, Jasper uses xerces if it is present.
grant codeBase "file:${catalina.home}/common/endorsed/xercesImpl.jar" {
  // Xerces needs to be able to read files in its Tomcat's home directory
  permission java.io.FilePermission "${catalina.home}${/}webapps${/}-", "read";
  permission java.io.FilePermission "${catalina.home}${/}work${/}-", "read";
  // Jasper needs to be able to read web application files
  permission java.io.FilePermission "C:${/}WebPublish${/}localhost${/}-", "read";
  // Xerces need to be able to read the jsp and web dtd's
  permission java.io.FilePermission
    "${catalina.home}${/}common${/}lib${/}jasper-runtime.jar", "read";
  // Xerces needs permision to access Tomcat utility resources
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.catalina.*";
  // Xerces needs permision to define Tomcat utility resources
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.catalina.*";
  // Xerces needs permision to access its jasper when compiling pages
  permission java.lang.RuntimePermission "accessClassInPackage.org.apache.jasper.*";
  permission java.lang.RuntimePermission "defineClassInPackage.org.apache.jasper.*";
  // Xerces needs this permission to start JMX for the admin application
  permission java.lang.RuntimePermission "createMBeanServer";
  permission java.lang.reflect.ReflectPermission "*";
};
```

```
// These permissions apply to the Xalan runtime if present in the
// common endorsed directory
grant codeBase "file:${catalina.home}/common/endorsed/xalan.jar" {
  // Xalan needs to be able to read files in its Tomcat's home directory
  permission java.io.FilePermission "${catalina.home}${/}webapps${/}-", "read";
  permission java.io.FilePermission "${catalina.home}${/}work${/}-", "read";
  // Jasper needs to be able to read web application files
  permission java.io.FilePermission "C:${/}WebPublish${/}localhost${/}-", "read";
  // Xalan need to be able to read the jsp and web dtd's
  permission java.io.FilePermission
    "${catalina.home}${/}common${/}lib${/}jasper-runtime.jar", "read";
};

// ========== WEB APPLICATION PERMISSIONS =====================================

// These permissions are granted by default to all web applications
// In addition, a web application will be given a read FilePermission
// for all files and directories in its document root.
grant {
  // Required for JNDI lookup of named JDBC DataSource's and
  // javamail named MimePart DataSource used to send mail
  permission java.util.PropertyPermission "java.home", "read";
  permission java.util.PropertyPermission "java.naming.*", "read";
  permission java.util.PropertyPermission "javax.sql.*", "read";
  permission java.util.PropertyPermission "javax.xml.*", "read";
  // Generic properties we allow to be read
  permission java.util.PropertyPermission "java.version", "read";
  permission java.util.PropertyPermission "java.vendor", "read";
  permission java.util.PropertyPermission "java.vendor.url", "read";
  permission java.util.PropertyPermission "user.dir", "read";
  permission java.util.PropertyPermission "file.encoding", "read";
  // OS Specific properties we allow to be read
  permission java.util.PropertyPermission "file.separator", "read";
  permission java.util.PropertyPermission "path.separator", "read";
  permission java.util.PropertyPermission "line.separator", "read";
  // Xerces and Xalan need this
  permission java.util.PropertyPermission "org.apache.xerces.*", "read";
  permission java.util.PropertyPermission "org.apache.xml.*", "read";
  // Xalan needs this
  permission java.lang.RuntimePermission "accessClassInPackage.sun";
  permission java.lang.RuntimePermission "accessClassInPackage.sun.*";
  // Needed for servlets and JSP's
  permission java.lang.RuntimePermission
    "accessClassInPackage.org.apache.catalina.util";
  permission java.lang.RuntimePermission
    "accessClassInPackage.org.apache.catalina.util.*";
  // XML Parsers need permision to access the jsp and web app dtd's
  permission java.lang.RuntimePermission
    "accessClassInPackage.org.apache.jasper.resources.*";
  // Needed by Xalan and Xerces
  permission java.io.FilePermission "${java.home}${/}lib${/}jaxp.properties", "read";
  permission java.io.FilePermission "${java.home}${/}lib${/}castor.properties", "read";
  permission java.io.FilePermission
    "${catalina.home}${/}common${/}endorsed${/}xalan.jar", "read";
  permission java.io.FilePermission
    "${catalina.home}${/}common${/}endorsed${/}xercesImpl.jar", "read";
  // Needed by struts
  permission java.util.PropertyPermission "jaxp.debug", "read";
  // Required for sending email
  permission java.io.FilePermission "${java.home}${/}lib${/}ext${/}mail.jar", "read";
```

```
  // Let all host names be resolved using DNS
  permission java.net.SocketPermission "*", "resolve";
  // Let all send email to a named SMTP host
  permission java.net.SocketPermission "replace-with-mail-server-name.domain:25",
"connect";
  // Permission to connect to the MySQL Database
  permission java.net.SocketPermission "localhost:3306", "connect";
};

// You can assign additional permissions to particular web applications by
// adding additional "grant" entries here, based on the code base for that
// application, /WEB-INF/classes/, or /WEB-INF/lib/ jar files.
//
// Different permissions can be granted to JSP pages, classes loaded from
// the /WEB-INF/classes/ directory, all jar files in the /WEB-INF/lib/
// directory, or even to individual jar files in the /WEB-INF/lib/ directory.
//
// For instance, assume that the standard "examples" application
// included a JDBC driver that needed to establish a network connection to the
// corresponding database and used the scrape taglib to get the weather from
// the NOAA web server.  You might create a "grant" entries like this:
//
// The permissions granted to the context root directory apply to JSP pages.
// grant codeBase "file:${catalina.home}/webapps/examples/-" {
//   permission java.net.SocketPermission "dbhost.mycompany.com:5432", "connect";
//   permission java.net.SocketPermission "*.noaa.gov:80", "connect";
// };
//
// The permissions granted to the context WEB-INF/classes directory
// grant codeBase "file:${catalina.home}/webapps/examples/WEB-INF/classes/-" {
// };
//
// The permission granted to your JDBC driver
// grant codeBase "file:${catalina.home}/webapps/examples/WEB-INF/lib/driver.jar" {
//   permission java.net.SocketPermission "dbhost.mycompany.com:5432", "connect";
// };
// The permission granted to the scrape taglib
// grant codeBase "file:${catalina.home}/webapps/examples/WEB-INF/lib/scrape.jar" {
//   permission java.net.SocketPermission "*.noaa.gov:80", "connect";
// };
```

Appendix D -- Sample MySQL Grants Tables

```
mysql> select * from user;
+-----------+------------+------------------+-------------+-------------+-------------+-------------
+---------
----+-----------+-------------+---------------+-------------+-----------+------------+-----------------
+-----
-------+------------+
| Host      | User       | Password         | Select_priv | Insert_priv | Update_priv | Delete_priv |
Create_p
riv | Drop_priv | Reload_priv | Shutdown_priv | Process_priv | File_priv | Grant_priv | References_priv |
Inde
x_priv | Alter_priv |
+-----------+------------+------------------+-------------+-------------+-------------+-------------
+---------
----+-----------+-------------+---------------+-------------+-----------+------------+-----------------
+-----
-------+------------+
| localhost | root       | xxxxxxxxxxxxxxxx | Y           | Y           | Y           | Y           | Y
    | Y         | Y           | Y             | Y           | Y           | Y          | Y
      | Y          |
| localhost | local      | xxxxxxxxxxxxxxxx | N           | N           | N           | N           | N
    | N         | N           | N             | N           | N           | N          | N
      | N          |
| localhost | localweb   | xxxxxxxxxxxxxxxx | N           | N           | N           | N           | N
    | N         | N           | N             | N           | N           | N          | N
      | N          |
| localhost | localrealm | xxxxxxxxxxxxxxxx | N           | N           | N           | N           | N
    | N         | N           | N             | N           | N           | N          | N
      | N          |
+-----------+------------+------------------+-------------+-------------+-------------+-------------
+---------
----+-----------+-------------+---------------+-------------+-----------+------------+-----------------
+-----
-------+------------+
4 rows in set (0.00 sec)
```

At the user level, grants are very strict!

User	Description
root	account for administering MySQL
local	account analogous to administrative account provided on production system
localweb	account analogous to account provided on production system for web app access
localrealm	account for Tomcat Manager authentication

```
mysql> select * from db;
+-----------+-------------+------------+-------------+-------------+-------------+-------------
+-------------+
-----------+------------+-----------------+------------+------------+
| Host      | Db          | User       | Select_priv | Insert_priv | Update_priv | Delete_priv | Create_priv
|
 Drop_priv | Grant_priv | References_priv | Index_priv | Alter_priv |
+-----------+-------------+------------+-------------+-------------+-------------+-------------
+-------------+
-----------+------------+-----------------+------------+------------+
| localhost | local       | local      | Y           | Y           | Y           | Y           | Y
|
 Y         | N          | Y               | Y          | Y          |
| localhost | local       | localweb   | Y           | Y           | Y           | Y           | N
|
 N         | N          | N               | N          | N          |
| localhost | local_realm | localrealm | Y           | Y           | Y           | Y           | N
|
 N         | N          | N               | N          | N          |
| localhost | local_realm | local      | Y           | Y           | Y           | Y           | Y
|
 Y         | N          | Y               | Y          | Y          |
+-----------+-------------+------------+-------------+-------------+-------------+-------------
+-------------+
-----------+------------+-----------------+------------+------------+
4 rows in set (0.00 sec)


At the db level, grants are more relaxed.

db           Description
local        Database for web application data
local_realm  Database for Tomcat Manager authentication
```

Appendix E -- Sample db Schema for Tomcat Manager Authentication

```
mysql> use local_realm
Database changed
mysql> show tables;
+-----------------------+
| Tables_in_local_realm |
+-----------------------+
| user_roles            |
| users                 |
+-----------------------+
2 rows in set (0.01 sec)

mysql> describe user_roles;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| user_name | varchar(80) |      | MUL |         |                |
| role_name | varchar(15) |      | MUL |         |                |
| rid       | int(10)     |      | PRI | NULL    | auto_increment |
+-----------+-------------+------+-----+---------+----------------+
3 rows in set (0.03 sec)

mysql> describe users;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| user_name | varchar(80) |      | PRI |         |                |
| user_pass | varchar(32) | YES  |     | NULL    |                |
| uid       | int(10)     |      | UNI | NULL    | auto_increment |
+-----------+-------------+------+-----+---------+----------------+
3 rows in set (0.03 sec)
```

This presenation has ended.
Click anywhere to return.

_

